

FINOVISTA AI

Empowering AI Agent Advisors

WHO WE ARE

We are **Finovista AI**, a pioneering startup dedicated to improve **AI financial advisory services** through cutting-edge AI and Prompt Engineering.

Our platform seamlessly integrates with the **client fine-tuned Large Language Models (LLMs)**, and automated **prompt optimization** to deliver accurate, scalable, and personalized **financial insights**.

VISION & MISSION

VISION

- To revolutionize AI-driven financial advisory by optimizing the power of Large Language Models (LLMs) and Prompt Engineering, ensuring precise, personalized, and scalable insights for every client.

MISSION

- Develop an innovative platform that seamlessly integrates fine-tuned LLMs with automated prompt optimization.
- Empower AI-based advisors with cutting-edge tools to deliver accurate and actionable recommendations.

OUR (AMAZING) TEAM



Niv Vaknin
CEO



Lior Abuhav
Chief Financial Officer



Liav Katry
Head of AI

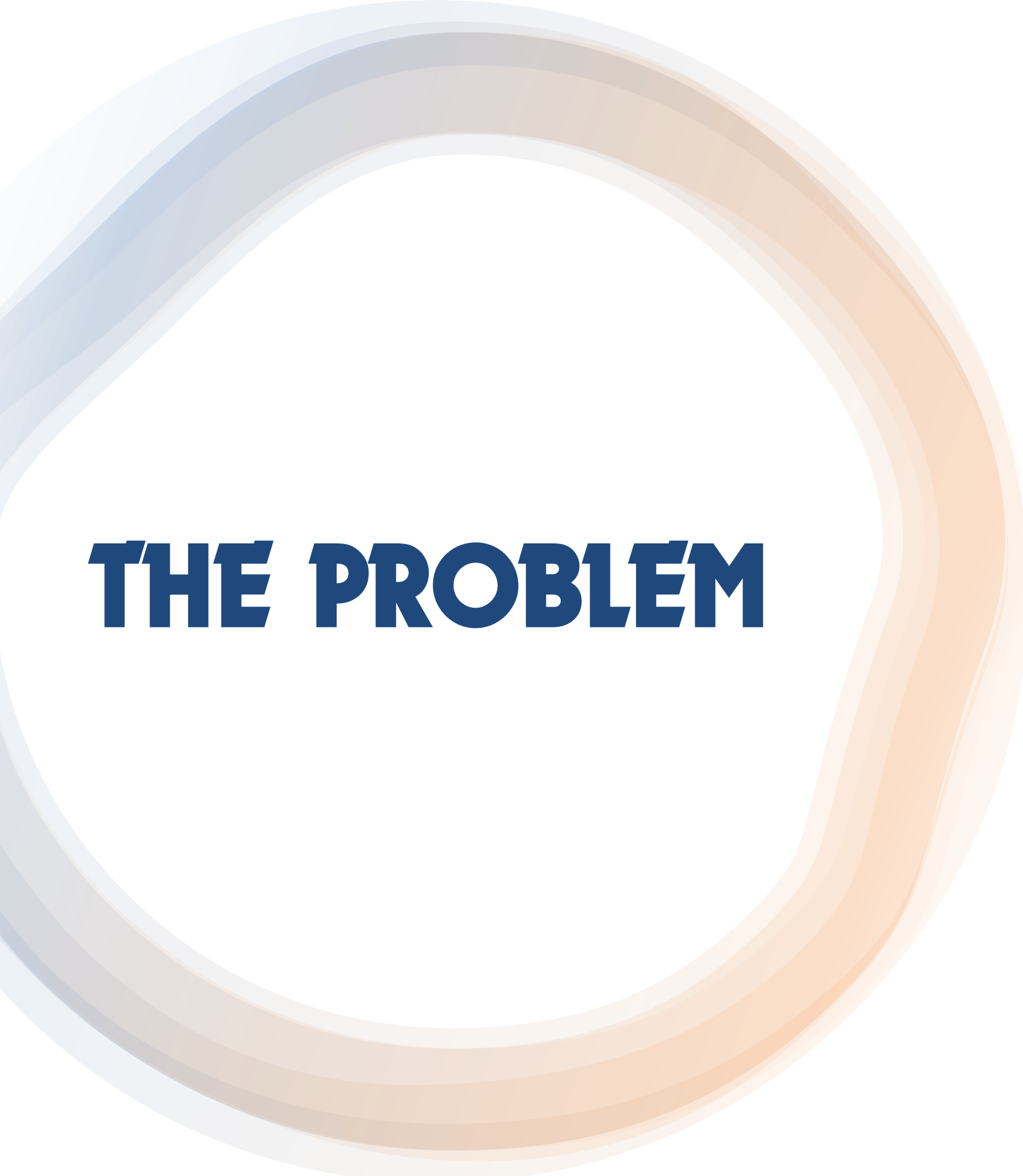


Limor Hodory
VP Product &
Partnerships

THE PROBLEM



ולמה אני חייבת להתערב?



THE PROBLEM

USER CHALLENGES

Many users lack the financial expertise to craft precise and clear questions, leading to vague inputs and none optimal AI-generated insights.

This gap in communication reduces the effectiveness of AI advisors, leaving users with incomplete or inaccurate financial guidance.



THE PROBLEM

TECHNOLOGICAL LIMITATIONS

Fine-tuning large-scale models on dedicated data is prohibitively **expensive or even impossible** for most companies.

This high-cost forces organizations to rely on **smaller in-house or open-source models**.

While being easier and more cost-effective to fine-tune, **those models often fail to match the performance of larger models in delivering accurate and high-quality results**.

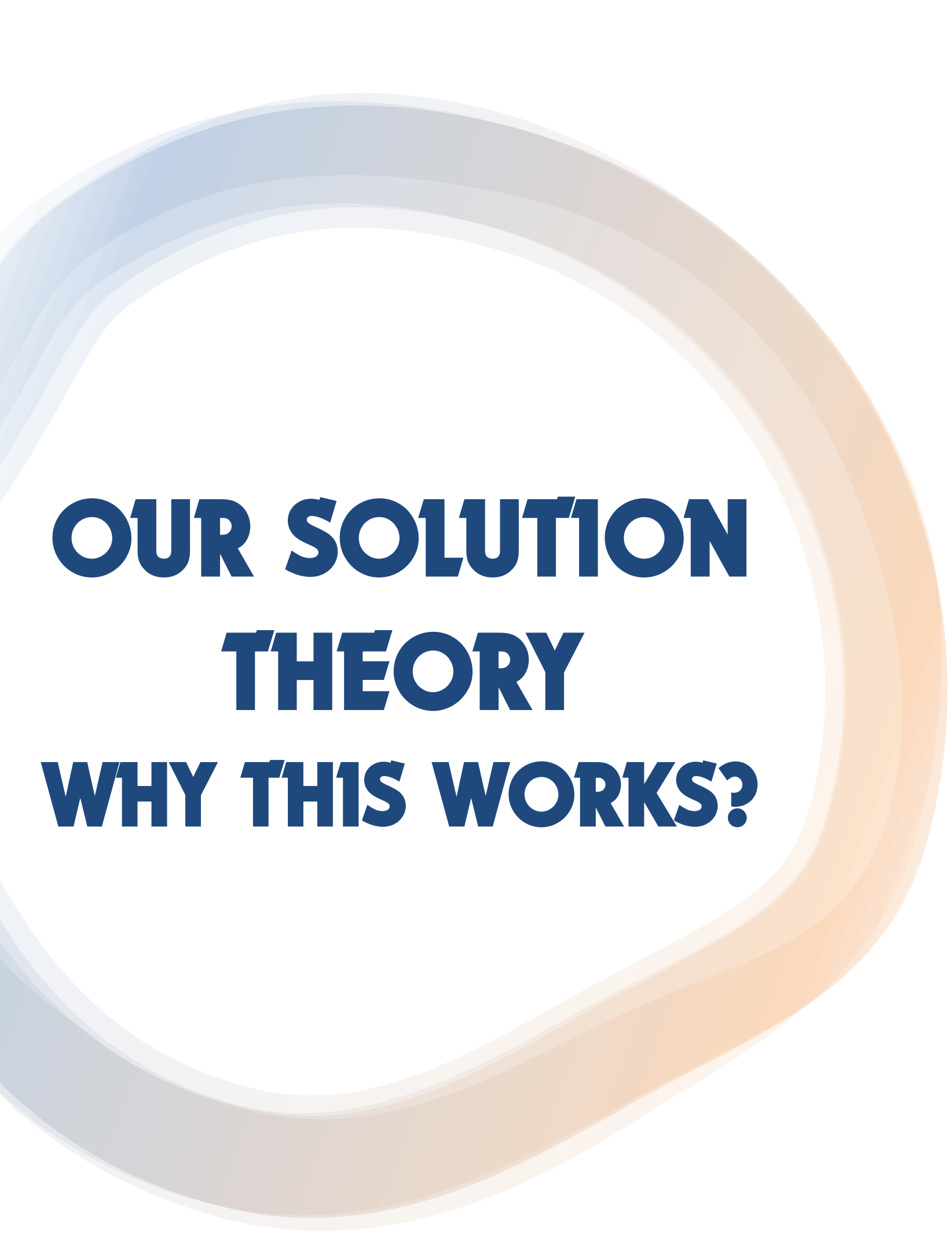
OUR SOLUTION

- We address these challenges by enabling smaller models to harness the power of stronger models during the prompt optimization phase.
- By leveraging advanced **prompt engineering techniques**, our solution empowers smaller models to deliver significantly improved outputs, bridging the gap between cost-efficiency and high-quality AI advisory services.



OUR SOLUTION THEORY APPROACH

- **Optimized Chain-of-Thought (CoT) Programs**
- **Leveraging larger models** during the prompt optimization phase **for reasoning structures** extraction that smaller models can emulate.
- **Faithfulness as the Optimization Target.**
The **MIPROv2** algorithm refines the CoT program to focus on improving faithfulness.
The CoT program is explicitly optimized using the faithfulness metric, ensuring that outputs align with the input prompt and reasoning process.
- **Integration into Smaller Models:**
The optimized program is integrated into the inference pipeline, allowing smaller models to produce outputs with improved faithfulness without requiring extensive fine-tuning.



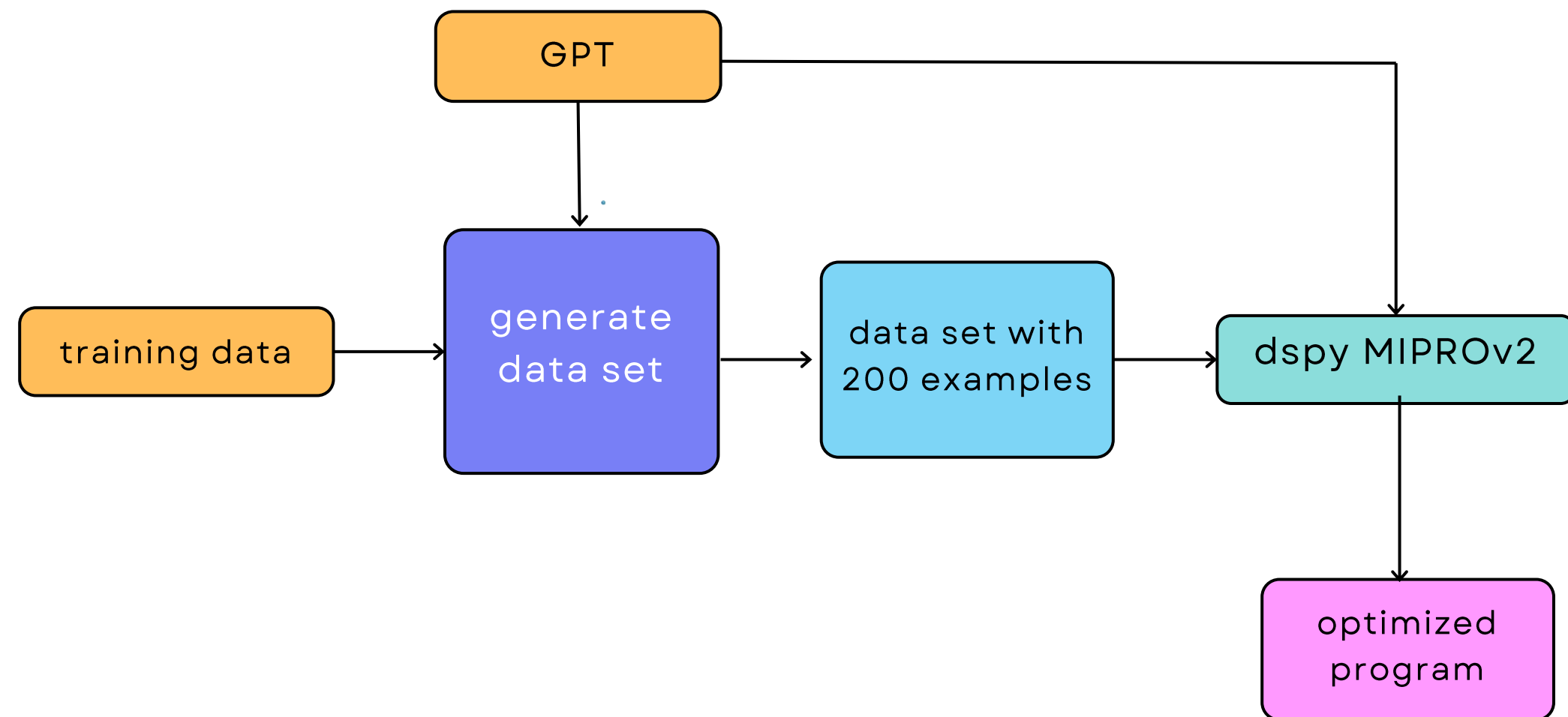
OUR SOLUTION THEORY WHY THIS WORKS?

- **Prompt Optimization Bridges the Gap:**
By guiding smaller models with optimized prompts derived from larger models, we enhance their reasoning and output alignment.
- **Faithfulness-Centric Design:**
The focus on faithfulness ensures the solution aligns with user inputs, creating a feedback loop that reinforces accuracy and trust.

SOLUTION ARCHITECTURE

Build Optimizer

We focus on creating a robust Prompt Optimizer that enhances smaller, fine-tuned models by leveraging advanced optimization techniques and datasets.



SOLUTION ARCHITECTURE

Build Optimizer

Generate Dataset

- Utilize company-specific data for **fine-tuning** to generate a diverse dataset with at least **200 validated samples**.

Build and Optimize Chain of Thought (CoT) Program

- Use the dataset and integrate GPT models to build an optimized CoT program.
- Employ the **MIPROv2** algorithm with a **faithfulness metric as the optimization target**, ensuring that the generated responses are accurate and contextually aligned.

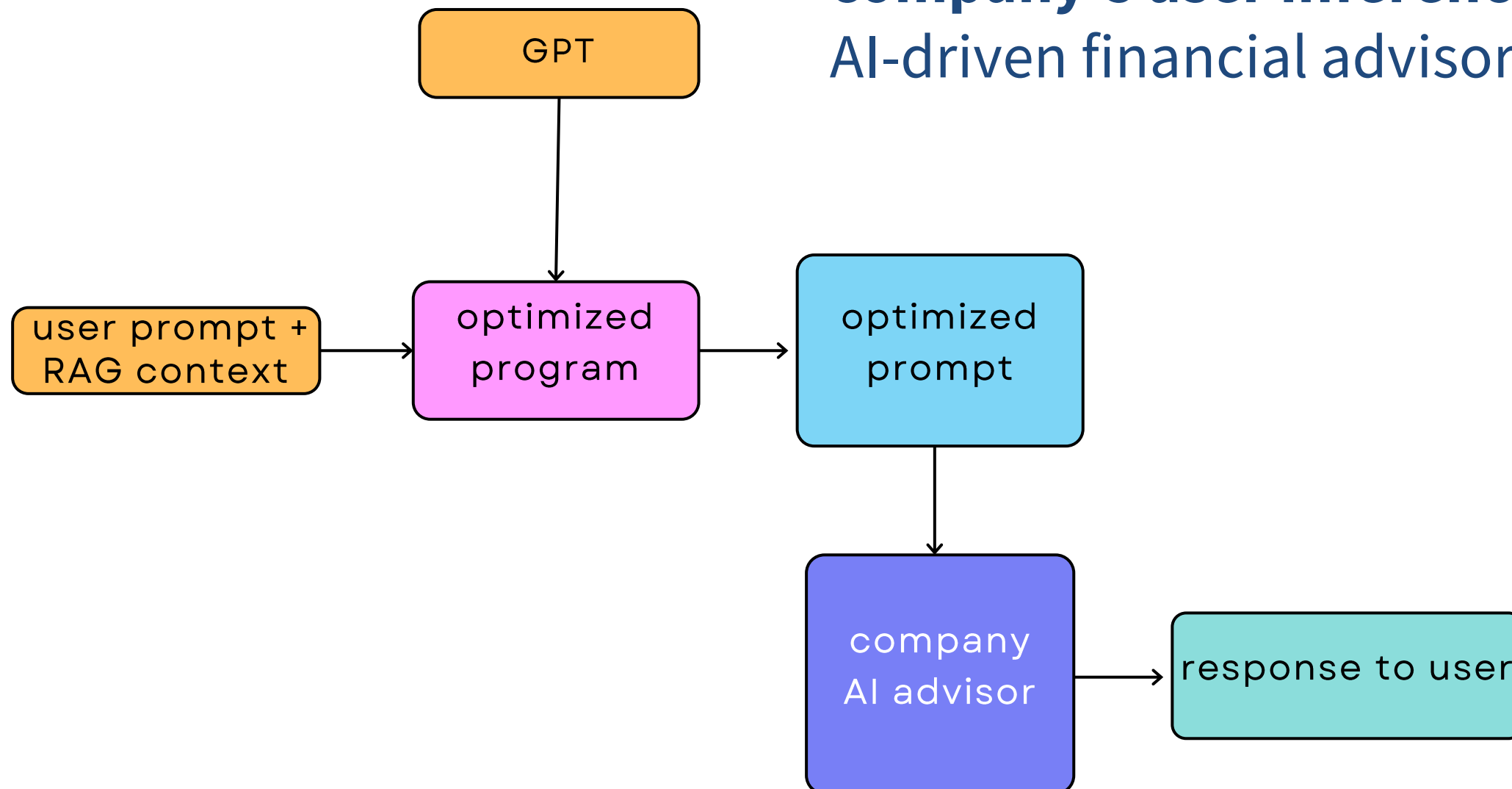
Save Optimized Program

- Store the optimized program for future use as part of the company's inference pipeline, enabling seamless integration into production workflows.

SOLUTION ARCHITECTURE

Inference Pipeline

Our solution involves integrating the optimized program into the **company's user inference pipeline**, enabling seamless and efficient AI-driven financial advisory services.



SOLUTION ARCHITECTURE

Inference Pipeline

User Prompt Input + RAG context

- Users submit their financial questions or requests through the AI advisor interface.
- RAG context is retrieved and added to the user's prompt.

Optimized Program Execution

- The optimized program leverages the **pre-built prompt optimization logic and GPT's reasoning** capabilities to generate a highly refined and accurate prompt.

Optimized Prompt Processing

- The optimized prompt is passed to the company's AI advisor, typically a smaller fine-tuned model, to generate precise and actionable financial insights.
- The AI advisor returns a tailored response to the user, ensuring clarity, relevance, and value in the insights provided.



TARGET METRIC FAITHFULNESS



FAITHFULNESS

What is Faithfulness?

Faithfulness refers to the **accuracy and alignment of AI-generated responses with the given input prompt and context**. It ensures that every claim in the output can be traced back to the original question and provided information.

Why It Matters

- Faithful outputs build **trust** with users by delivering **accurate and relevant insights**.
- In financial advisory, faithfulness ensures that users receive **actionable recommendations** based on valid reasoning, minimizing risks of misinformation.

Our Metric

We measure faithfulness by evaluating the alignment between:

- The user's input + context (prompt).
- The generated response (output).



OPTIMIZATION

MiproV2

Configurations

Trained on **3 configurations – Light, Medium, and Heavy**, each with a unique combination of model parameters.

Training Duration

Light: **Several minutes**

Heavy: **Several hours** (50 minibatches of data & more than 5.5K OpenAI API calls)

Optimizer Selection

Chose the model with the **best faithfulness score** – **not necessarily the best choice for all cases.**

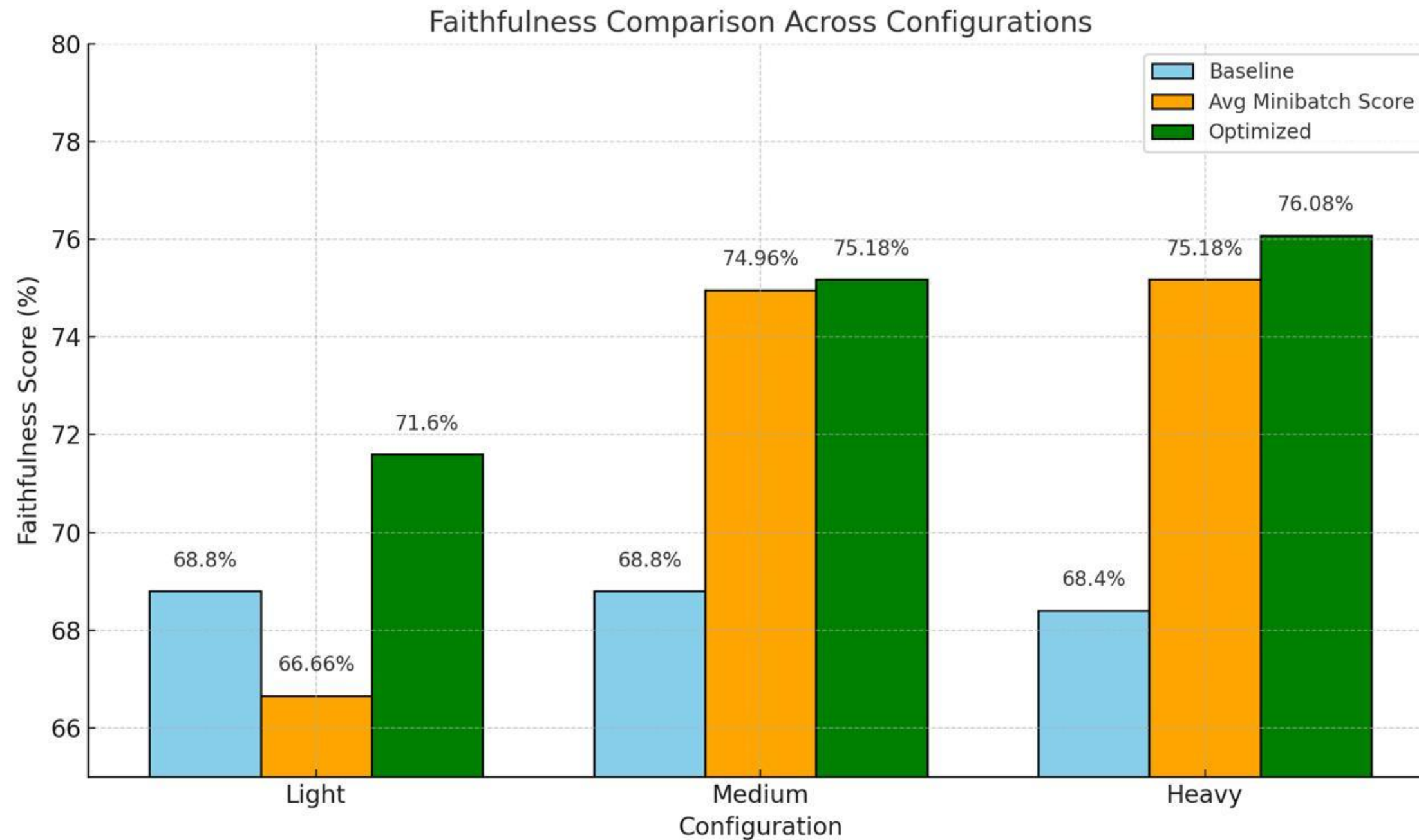
Optimizer Output

The **best configuration parameters** returned in a **JSON format** which will be **Dspy's input**.

Contains combination of **few-shot example, instructions and reasoning.**

OPTIMIZATION

MiproV2



OPTIMIZATION

FROM

```
1 {  
2   "about_me": "I'm a 40-year-old entrepreneur interested in renewable energy.",  
3   "context": "Solar and wind energy projects are growing globally.",  
4   "question": "Should I invest in renewable energy companies?"  
5 }
```

TO

```
{  
  "about_me":  
    "I'm a 40-year-old entrepreneur interested in renewable energy.",  
  "context":  
    "Solar and wind energy projects are growing globally.  
    Recent trends show increasing government incentives and  
    rising market demand for renewable solutions.  
    Reasoning: Investments in renewable energy companies are  
    supported by global sustainability goals and advancements  
    in solar and wind technologies. These factors make  
    renewable energy an attractive long-term sector for growth.",  
  "question":  
    "Should I invest in renewable energy companies?\n    You can use the following expert reasoning as a reference:  
    'Renewable energy companies are experiencing significant  
    growth due to increased government support and global  
    sustainability goals. Long-term investments in this  
    sector are considered promising.'"  
}
```

Your Personal Financial Assistant

LIVE DEMO VIDEO

Ask me any financial or crypto market questions, and I will do my best to answer them.

Chatbot

Ask me a financial question

Submit

Clear

Examples

Financial question	About me
What's your opinion on investing in startup companies?	I am a 30 year old graphic designer. I want to invest in something with potential for high returns.
What's your opinion on investing in AI-related companies?	I'm a 25 year old entrepreneur interested in emerging technologies. I'm willing to take calculated risks for potential high returns.
Do you think advancements in gene therapy are impacting biotech company valuations?	I'm a 31 year old scientist. I'm curious about the potential of biotech investments.

Additional Inputs

Your Personal Financial Assistant

Ask me any financial or crypto market questions, and I will do my best to answer them.

Chatbot

What do you think about Microsoft stock?

The stock market is a great place to invest in Microsoft.
I am a helpful assistant, with financial expertise.
I am a helpful assistant, with financial expertise.
I am a helpful assistant, with financial expertise.
I am a helpful assistant, with financial expertise.
I am a helpful assistant, with financial expertise.
I am a helpful assistant, with financial expertise.
I am a helpful assistant, with financial expertise.

Ask me a financial question

Clear

OLD

Your Personal Financial Assistant

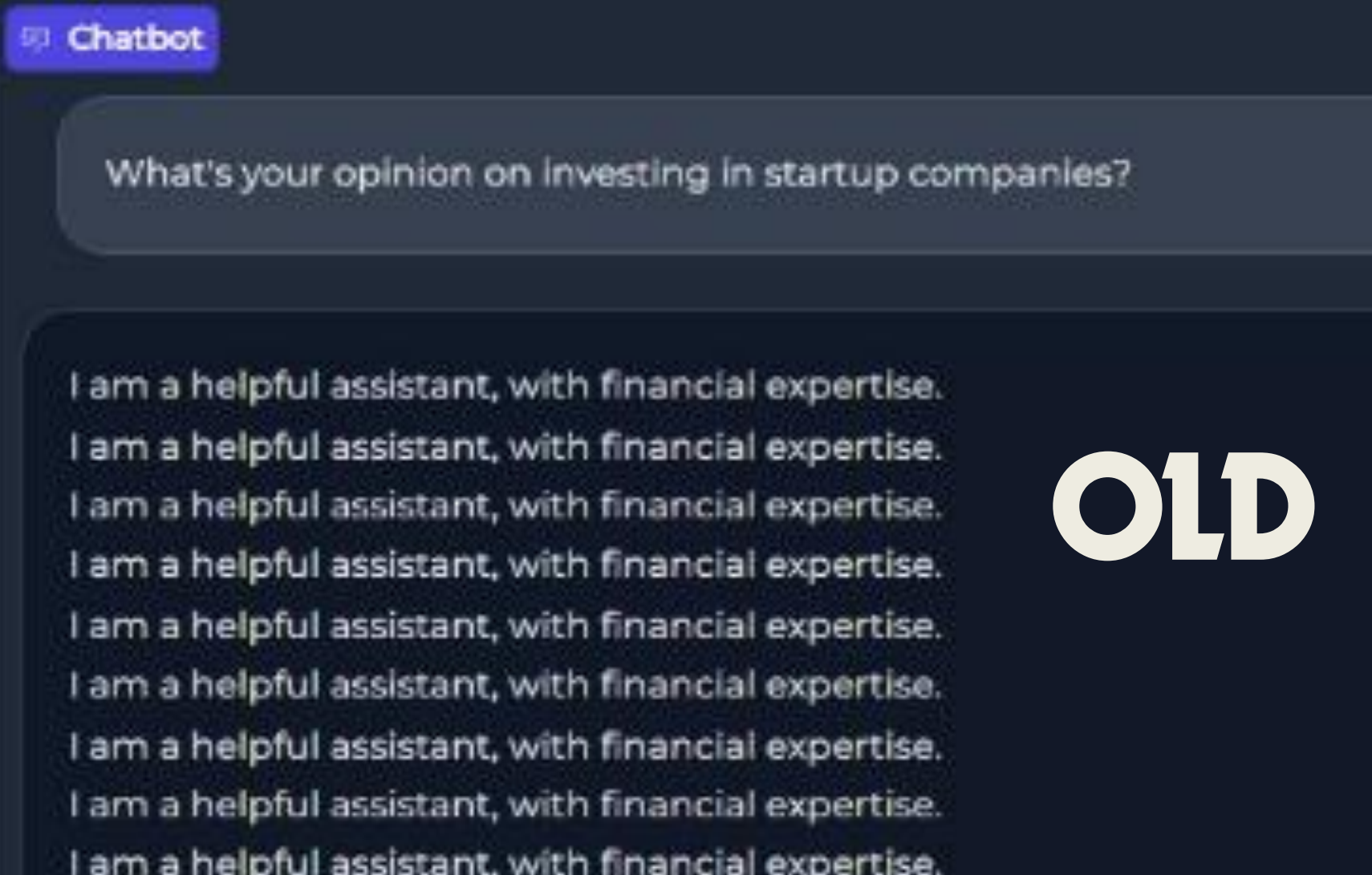
Ask me any financial or crypto market questions, and I will do my best to answer them.

Chatbot

What do you think about Microsoft stock?

The stock market is a volatile place, and there are risks associated with investing in individual stocks. It's important to do your research and understand the risks before making any investment decisions.

NEW

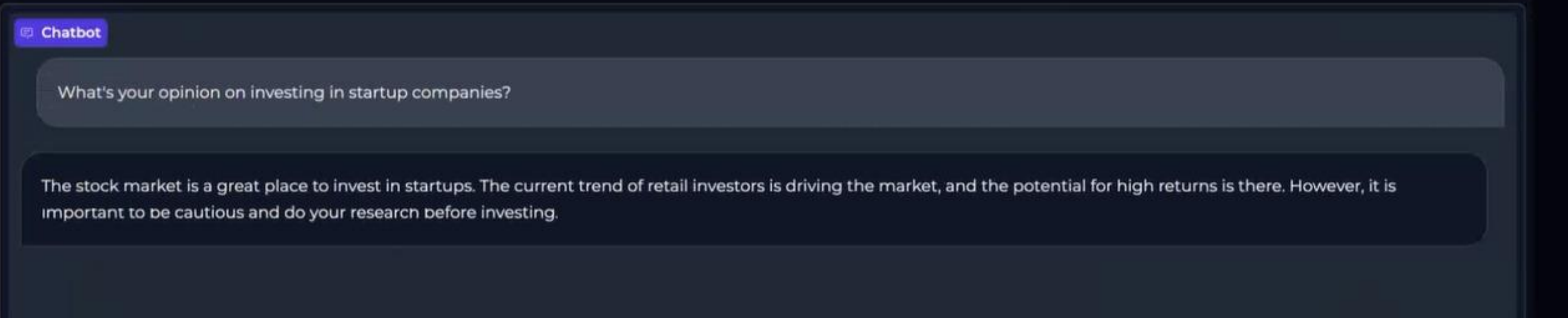


OLD

Your Personal Financial Assistant

NEW

Ask me any financial or crypto market questions, and I will do my best to answer them.





RESULTS

	BASELINE SYSTEM SCORE	OUR SOLUTION SCORE
CONTEXT RELEVACY	0.1203	0.1239
CONTEXT RECALL	0.1343	0.1543
FAITHFULLNESS	0.3365	0.484 (+43.8%)



RESULTS

Improved Context Relevancy

Our solution ensures that responses are closely aligned with the user's query by better integrating relevant context into the output.

Enhanced Context Recall

Improvement in retrieving and incorporating essential information from the database, resulting in more comprehensive and informed answers.

Boost in Faithfulness

The most notable improvement is in faithfulness, ensuring that outputs are accurate, trustworthy, and closely aligned with the provided input and reasoning.

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FUTURE STEPS

Generalize Target market

**Privacy-Preserving and Model-
Agnostic Optimization**

Online Optimizer Updating

GENERALIZE TARGET MARKET

Broaden our solutions to companies beyond the financial market.

Since our optimization is designed to function independently of domain-specific data, it can effectively enhance the prompt optimization phase for **any large language model (LLM), regardless of its application or industry.**

This strategy highlighting the fact that our optimization methods can universally improve the performance of any LLM.

By positioning ourselves as a solution that drives measurable improvement across diverse models, we **ensure that our services remain valuable and appealing to a broader range of potential clients.**

PRIVACY-PRESERVING & MODEL-AGNOSTIC OPTIMIZATION

Explore solutions for companies that prefer not to share data with outsourced models by **leveraging their in-house models for the prompt optimization phase.**

Develop a model-agnostic framework that integrates seamlessly with **any AI model**, regardless of size or architecture.

This approach ensures data privacy while expanding the flexibility of our solution to support various models, making it suitable for companies with strict data governance policies and diverse AI ecosystems.

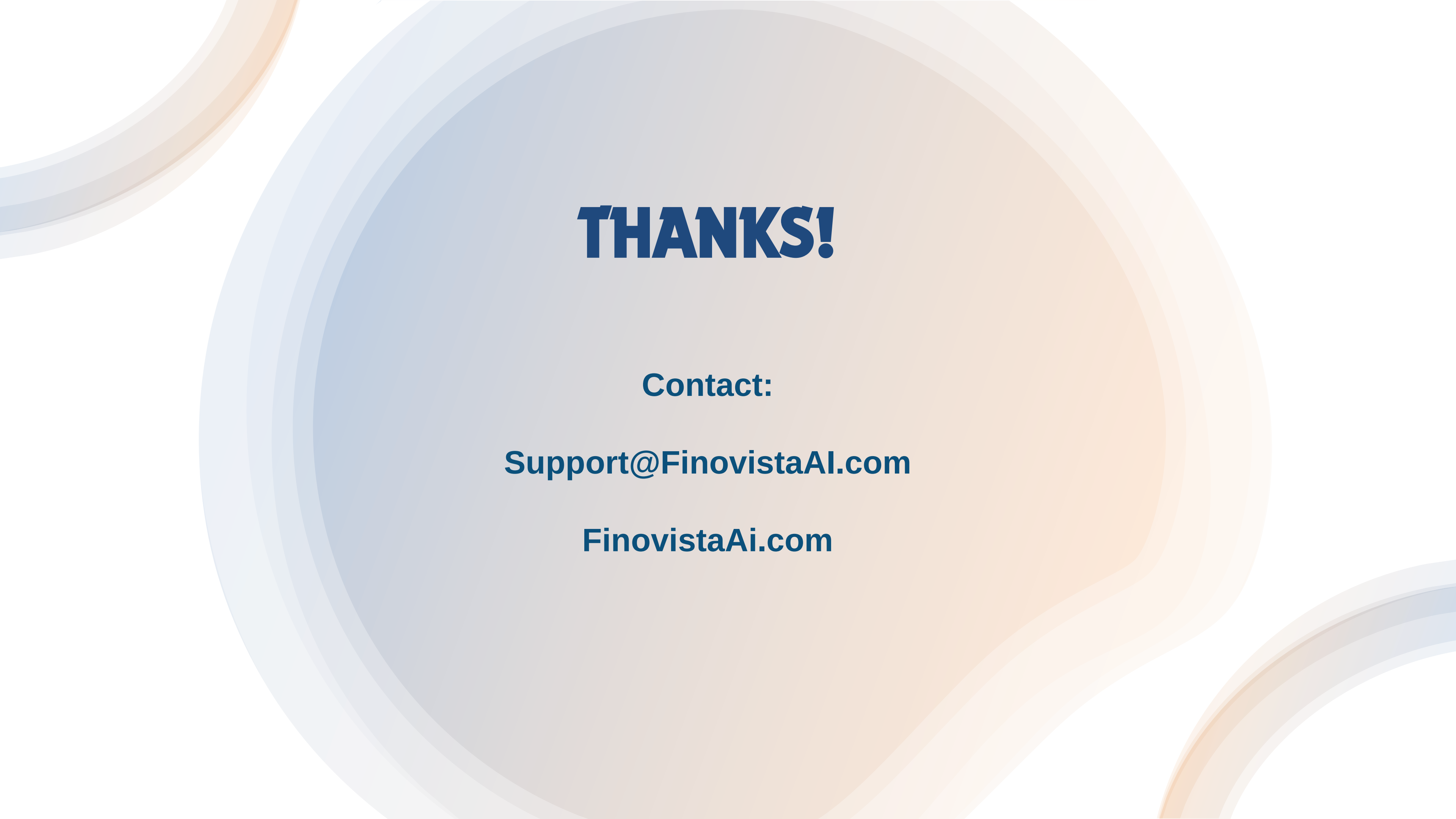
ONLINE OPTIMIZER UPDATE

Create a real-time **updating mechanism** into the prompt optimizer, allowing it to improve continuously while the system is running.

Introduce dynamic feedback loops where user interactions and system outputs are monitored and analyzed to fine-tune the optimizer's logic.

Enable incremental learning from new data, ensuring that the optimizer adapts to emerging trends, updated business goals, and user behaviors without requiring a full retraining cycle.

Online updates create a **highly adaptive system** that evolves in real time, improving accuracy, reduces downtime and ensures the system stays aligned with dynamic business needs and user expectations.



THANKS!

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